

Stats 101C Kaggle Competition

Classification of Cancerous Cells

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Introduction

1

Problem

2

About the dataset

Problem

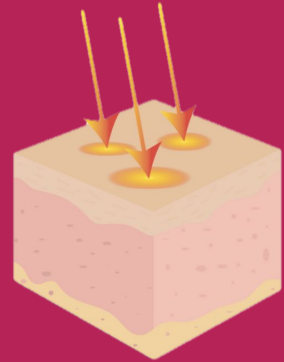


5 million+ cases diagnosed annually

1 in 5 Americans will develop skin cancer by age 70

Occur when DNA damage causes skin cells to grow uncontrollably

- Most often in sun-exposed areas
- Form malignant tumors in the skin's layers



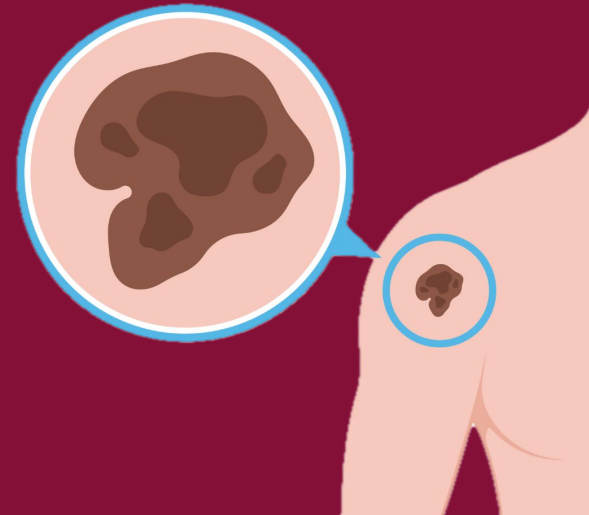
Early detection

→ improving treatment outcomes and survival rates

1. Analysis of predictor
and their relation to risks of skin cancer
2. Classification of the "Cancer" status of an individual
 - ; **50,000** observations training data set
 - ; **20,000** observations testing data set

Benign	Malignant
23868	26132

Predicting Skin Cancer Status



with 49
predictors

Demographic

Biological & Health

Sun Protection & Skincare

Environmental & Exposure

Behavioral, Lifestyle & Miscellaneous

Methodology

1

Data Cleaning

2

Feature Selection

3

Modelling



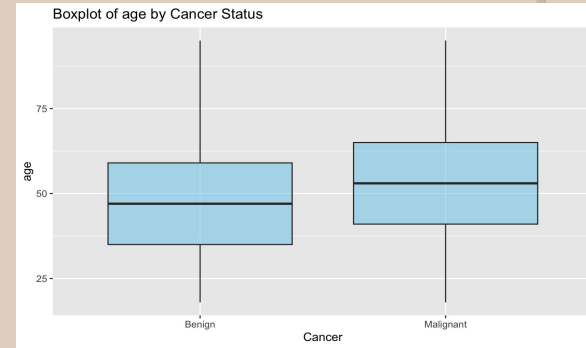
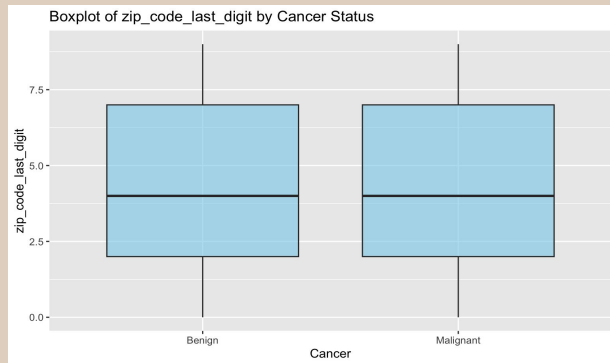
Data Cleaning

age	gender	skin_tone
3930	4037	4091
education	income	urban_rural
3972	4058	3998
occupation	avg_daily_uv	sunscreen_freq
3909	4054	4075
sunscreen_spf	sunscreen_type	sunscreen_brand
4085	3979	4000
hat_use	clothing_protection	tanning_bed_use
3984	3966	4030

- With **196,042** missing values, they were spread out in each column.
- Each column had nearly **4,000** missing values each.
- Total of **7%** of the dataset was missing values.
- In order to deal with imputing these missing values and doing further feature selection from **49** predictors given, we did some exploratory data analysis.

Exploratory Data Analysis

With missing values, we wanted to check which numerical values had the most significant differences in their medians for malignant vs. benign lesions, so we plotted boxplots for each predictor



Thus, we found predictors with almost no difference between benign and malignant medians like alcohol drinks per week, frequency_doctor_visits_per_year, exercise_freq_per_week, commute_minutes, zip_code_last_digit, monthly_screen_time_minutes, so we further analyzed them using chi-sq test and one-way anova

Dealing with predictors

Numerical Predictors:

- Ran summary statistics for all the numerical variables.
- Predictors like **BMI** and **Income** have numerical values but do not provide more information than the whether a person is **underweight, normal, overweight or obese** for BMI and **low, middle and high** for income.
- Converted them to categorical predictors as factors.

Categorical Predictors:

- Found levels of all the categorical predictors and grouped them to fewer levels
- Reduced factor levels of **skin_tone** and **lesion_color** from 6 to 3 each.
- Predictor **outdoor_job** had only 2 values (0 & 1) inputted as numerals so converted the predictor to **categorical with 2 factors**.

Dealing with predictors

To find the most important categorical predictors, we conducted a **chi-squared test** between the categorical predictors and Cancer response variable.

family_history	skin_tone	sunscreen_freq	immunosuppressed
9.781167e-118	5.433568e-80	9.786698e-49	4.517045e-41
outdoor_job	occupation	tanning_bed_use	clothing_protection
2.133525e-24	1.241510e-19	1.732939e-18	6.437897e-15
skin_photosensitivity	hat_use	urban_rural	income
1.053908e-12	5.860108e-10	1.015537e-01	1.110831e-01
participates_outdoor_sports	uses_smartwatch	music_genre	
1.753640e-01	2.096736e-01	2.372982e-01	

Before dealing with NA's we can see family_history, skin_tone, sunscreen_freq, immunosuppressed, outdoor_job, occupation, tanning_bed_use, clothing_protection, skin_photosensitivity and hat_use are some of the most important variables as they have the lowest p-values (<0.05)

Dealing with predictors

To find the most important numerical predictors, we performed a **a one-way ANOVA** between the numerical predictors and Cancer response variable.

Before dealing with NAs we can see **age**, **avg_daily_uv**, **number_of_lessons**, **sunburns_last_year**, **lesion_size_mm**, **sunscreen_spf**, **years_lived_at_address** are some of the most important variables as they have the lowest p-values (<0.05)

age 1.022329e-203	avg_daily_uv 5.377946e-41
number_of_lessons 2.998532e-32	sunburns_last_year 1.199340e-26
Lesion_size_mm 5.412025e-07	sunscreen_spf 1.038599e-04
Years_lived_at_address 2.665885e-02	desk_height_cm 6.473242e-02
Residence_lon 1.079351e-01	exercise_freq_per_week 2.068466e-01
Zip_code_last_digit 2.998131e-01	distance_from_beach_km 4.021359e-01
Alcohol_drinks_per_week 5.972993e-01	commute_minutes 7.943421e-01

Final predictors

- family_history
- skin_tone
- sunscreen_freq
- immunosuppressed
- outdoor_job
- occupation
- tanning_bed_use
- clothing_protection
- skin_photosensitivity
- hat_use
- age
- avg_daily_uv
- number_of_lesions
- sunburns_last_year
- lesion_size_mm
- sunscreen_spf
- years_lived_at_address

17
Predictors

Dealing with NAs

The final 17 predictors that had missing values were imputed using various methods and their best results are as follows:

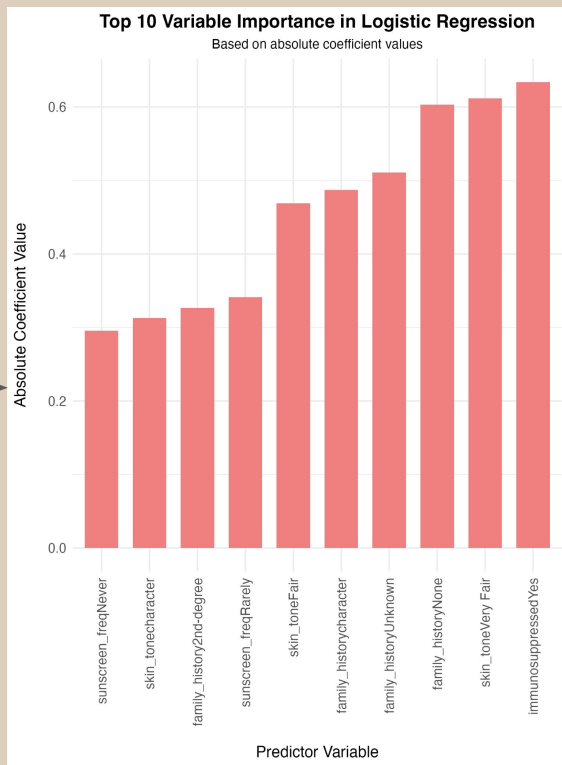
Method	Results	Model
Median/Mode	0.59375	Logistic
Hot Deck	0.60170	LDA
MissForest	0.60360	LDA
MissForest	0.60415	Logistic
MICE	0.60080	Logistic
Denoising Autoencoder	0.5911	Logistic

Logistic Regression

**Full model with all
predictors**

Accuracy score of
0.60475

With 10-fold cross validation



**Reduced model with only
significant predictors**

Accuracy score of
0.60080

**Reduced model with our 15
predictors**

Accuracy score of 0.60055

Lasso Regression

- MissForest imputation performed best with Lasso with max iterations = 2 and number of trees = 20
- Lasso performed best on the reduced model of 15 chosen predictors
- Sunscreen_spf and occupation were removed due to their insignificance in logistic regression
- **No variable selection was done by Lasso**

Accuracy	0.6004
Best λ	0.000488865
Threshold (ROC)	0.5375

Random Forest

- **Basic** Random Forest model **Accuracy** using **mtry 7, ntree 1,000**: 0.5854

on kaggle: 0.5963

Confusion Matrix

		Predicted	
		Benign	Malignant
Actual	Benign	11865	12003
	Malignant	8728	17404

- **Best mtry** Random Forest model **Accuracy** using **mtry 10** : 0.5899

on kaggle: 0.5930

```
best_mtry <- tuneRF(train[model_cols], train$Cancer, stepFactor=1.5, improve=0.01)
```

- RandomForest function-based imputation showed no significant improvements

KNN and Naive Bayes

KNN Accuracy using
MissForest on Cross
Validation of training data
(k's tried from 3 to 25):
0.6191

Naive Bayes using
MissForest on
cross-validation of
training data:
0.59745

Naive Bayes using
median/mode imputation
on kaggle:
0.59745

		Predicted	
		Benign	Malignant
Actual	Benign	14650	9825
	Malignant	9218	16307

LDA and QDA

Various models were tested, with the top and bottom performing listed below:

- QDA with Missforest
- LDA with Eigen Shrinkage
- LDA with Missforest (Full Data Set)
- LDA with HotDeck (Full Data Set)
- LDA with Missforest (Chosen 15 Variables)

Model	Accuracy (Testing)	Accuracy (Training)
QDA w/ MF	0.57560	0.56996
LDA w/ ES	N/A	0.47736
LDA w/ MF (Full)	0.6036	0.59712
LDA w/ HotDeck (full)	0.6017	0.59376
LDA w/ Missforest (chosen 15)	N/A	0.5984

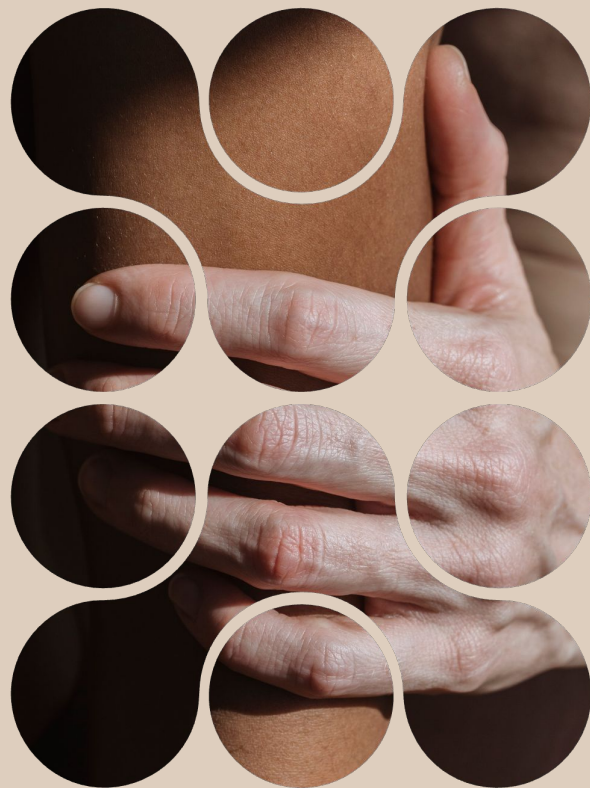
GAM

```
> gam(formula = Cancer ~ s(age, df = 4) + s(avg_daily_uv, df = 4) + s(lesion_size_mm, df = 4)  
+   gender + skin_tone + family_history, family = binomial, data = train.data)
```

Confusion Matrix

		Predicted	
		Benign	Malignant
Actual	Benign	12177	11691
	Malignant	9130	17002

GAM Accuracy using
median/mode imputation:
0.5842



Result & Discussion

① Analysis

② Summary

② Discussion

Model Summary

Logistic Regression

KNN and Naive Bayes

Lasso Regression

LDA and QDA

Random Forest

GAM



Performed with best accuracy

Logistic Regression

LDA

Analysis: Key Takeaway

Model

Logistic Regression
10-fold CV

Observations

50000

Predictors

49

Simplicity

Complex

Accuracy

0.60475

Rank

Top 30

Analysis: Key Takeaway

Model

LDA LOOCV

Observations

50000

Predictors

49

Simplicity

Complex

Accuracy

0.60360

Rank

Top 30

Analysis: Key Takeaway

Model

Logistic Regression
10-fold CV

Observations

50000

Predictors

15

Simplicity

Simple

Accuracy

0.60055

Rank

Top 30

Logistic Regression vs LDA

Logistic Regression

- Simpler model - reduced complexity leads to higher bias but lower variance
 - Can adapt to the testing data better
- Linearity assumption
- No assumptions about covariance

Linear Discriminant Analysis

- More complex - high variance
- Strong assumptions about the data (covariance is equal across classes)

Discussion

The Top Predictors and Models

- The 15 predictors retained from the original set of 17 represent the optimal model
- Removing any of them decreases accuracy

Logistic 49 Logistic 15 LDA 49 LDA 15 KNN 17

- Top performing models were logistic regression and LDA
- However, there are more models we would like to explore, as we would like a higher accuracy, as all models are performing relatively similar
- KNN showed promising results on the training data to increase accuracy
- ANNs may be able to bring the accuracy higher, though they are more complex

Limitations & Conclusion

1

Limitations

2

Conclusion



Limitations

Missing Values: Imputing missing values using methods like MissForest, KNN, Autoencoder may perform well but still impute model-based values



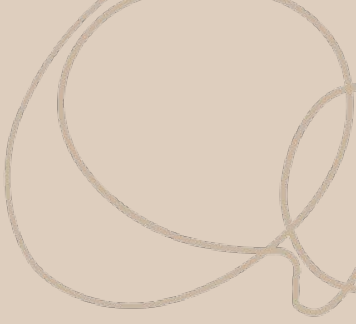
Computationally expensive: imputing and modelling on 50000 rows & 50 predictors is computationally expensive and slow



Model Assumptions: Each model has structural assumptions and may not match the true complexity of medical data



Conclusion



Overall, logistic regression is preferable of the two due to its simplicity and less strong assumptions. However, all models performed very similarly, indicating that another classification model might be necessary. For next steps, we'd like to look at more complex models such as ANNs.

